



RESEARCH ARTICLE

VocalCoach: A Single-Backbone Multi-Task Model for Robust Offline Singing Analysis and Coaching

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Abstract

VocalCoach is an offline singing-analysis system that turns one vocal recording into structured coaching feedback. From a single forward pass of a 7.8M-parameter temporal-convolutional backbone, it produces pitch (F0), voice activity, singing technique, note onsets/offsets, and a quality score, rendered as a browser coaching report. The central problem is robustness: models trained on clean studio audio degrade on real-world recordings made with consumer microphones. On-the-fly gain and spectral-tilt augmentation help close that gap, lifting out-of-distribution voiced-detection rate from ~53% to 68% while holding pitch accuracy at 99% in-distribution (72% out-of-distribution). Technique reaches 80.8% clip accuracy (matching published VocalSet state-of-the-art) and the note head reaches 0.81 onset/offset F1. The key result is methodological: each capability is best served by a *VAD-preserving probe* on a shared, augmentation-hardened backbone rather than a single jointly-trained model. Several plausible additions (a falsetto specialist, a nine-dimension quality head) were measured, found not to separate, and dropped.

Keywords: singing voice analysis, multi-task learning, pitch estimation, singing technique, note transcription, domain robustness

1. Goal and Motivation

Vocal coach. VocalCoach is an automated singing tutor: a singer uploads a recording and receives the feedback a human vocal coach would give — what pitch was sung, whether notes were stable, which techniques were used (vibrato, breathiness, belting), where the phrases were, and an overall quality assessment — with an optional reference recording to compare against.

Target user and problem. The target user is an amateur or a journeyman singer practising alone, without access to a coach. The concrete problem is that existing tools are either pitch-only tuners that say nothing about technique or musical quality, or cloud services that assume studio-quality input and fail on real-world conditions. Real practice audio can be quiet, noisy, and affected by inexpensive microphones — exactly the scenario where research models, trained on clean audio, break down.

System goal. *Produce a complete, technique-aware coaching report from a single real-world singing clip, robustly enough to work on amateur recordings, in one offline pass of one shared model.*

2. Related Work

Singing analysis spans several sub-problems usually solved separately; our contribution is unifying them on one robust backbone and being disciplined about which capabilities hold up.

Pitch estimation. CREPE Kim et al. (2018) and RMVPE Wei et al. (2023) frame F0 estimation as classification over a binned posteriorgram. We adopt this formulation (360 bins, 20-cent resolution) and use RMVPE to generate training labels, but unlike these single-task models we share the pitch representation with four other heads and add an offline Viterbi decode for clean F0 tracks.

Singing-technique classification. Recent classifiers on VocalSet (Wilkins et al., 2018) report ~80% clip accuracy on its technique classes. We match this (80.8%) but show that the commonly-paired GTSinger falsetto data (Zhang et al., 2024) does not add a usable fifth class once measured properly.

Note transcription. Frame-level onset/offset detection (Hawthorne et al., 2018) combined with a pitch posteriorgram yields discrete notes. Our note head follows this design; our finding is that the *pitch-at-note* ceiling is set by the posteriorgram’s octave errors, not the onset detector.

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Singing quality. SingMOS (SingMOS contributors, 2024) provides scalar mean-opinion scores. We distil a scalar quality head from it with a contrastive pro-versus-amateur ranking term, and show that multi-dimensional expert ratings we examined are too inter-correlated to learn as independent axes.

Our augmentation pipeline follows SpecAugment (Park et al., 2019) and extends it with gain and spectral-tilt perturbations targeting the loudness and microphone-colour gap specific to amateur recording setups.

3. System Overview

A single log-mel input (40 bands, 16 kHz) passes through one shared backbone — 8 dilated temporal-convolution blocks plus 4 self-attention layers (hidden = 256) — to produce frame features of shape $(B, T, 256)$. From these, five lightweight heads read off all outputs in one forward pass, as summarised in Table 1 and Figure 1.

Four of the heads are per-frame, but the quality head *mean-pools the frame features over time first* ($\mathbf{x}.\text{mean}(\text{dim} = 1)$) and emits a single clip-level score — it does not see per-frame features. The browser UI consumes this report to draw the waveform with voice activity, the log-mel spectrogram, the pitch posteriorgram, a note timeline, a per-phrase breakdown (with clip-segment playback), the detected technique, and the overall score. An optional reference recording is analysed in the same pass for an A/B comparison with synced playback.

Head	Granularity	Output
Pitch	per-frame	360-bin posteriorgram, F0
VAD	per-frame	voiced / unvoiced
Technique	per-frame	4 classes
Note	per-frame	onset/offset, MIDI
Quality	clip-level	one 0–100 score

Table 1: The five heads and their output granularity. Technique is per-frame in the model but aggregated over voiced frames for display; quality is clip-level by construction (time-pooled).

3.1 Demonstration

Figures 2 and 3 show the running system on a real held-out clip (Vocadito). These are qualitative illustrations of the interface and the coaching output, not quantitative evidence; the measured results are reported in Section 6. Figure 2 shows the top-level analysis — overall score, waveform with voice activity, log-mel spectrogram, and pitch posteriorgram — while Figure 3 shows the per-phrase coaching breakdown, where each detected phrase carries its note, dynamics, and a technique-specific comment with an actionable suggestion.

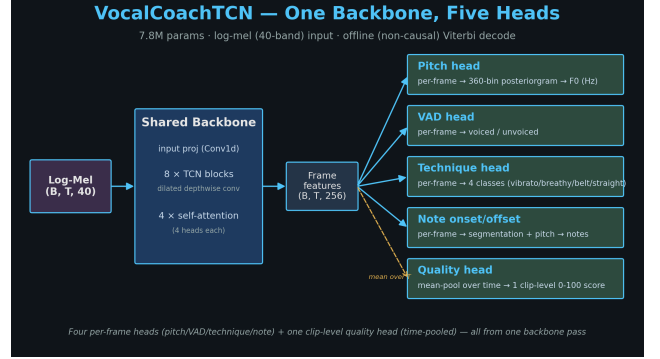


Figure 1: VocalCoachTCN: one shared backbone feeds four per-frame heads (pitch, VAD, technique, note) and one clip-level quality head that mean-pools the frame features over time.



Figure 2: The analysis view on a real clip: overall score, waveform/voice activity, log-mel spectrogram, and pitch posteriorgram (Qualitative demonstration only).

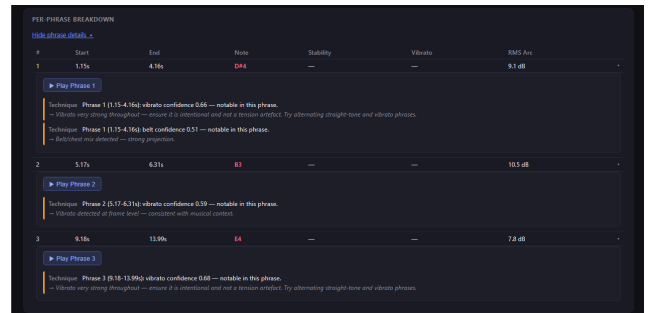


Figure 3: Per-phrase coaching breakdown: each phrase lists its note, RMS arc, and a technique-specific observation with an actionable tip, with per-phrase audio playback (Qualitative demonstration only).

4. Data

Each capability is supervised by the dataset best suited to it (Table 2); all are public and linked rather than redistributed.

Use	Dataset	Size
Pitch / VAD	Merged P-VAD	51,627 clips, 126 h
Technique	VocalSet	824 clips
Note	Annot. VocalSet	824 clips, 4.5k notes
Quality	SingMOS+pairs	MOS, pro/amateur
OOD eval	Vocadito	40 real recordings

Table 2: Datasets, one per capability. Vocadito is held entirely out of training as the out-of-distribution (OOD) generalisation test.

Provenance and preprocessing. All audio is re-sampled to 16 kHz mono; features are log-mel (40 bands) plus RMVPE-derived F0 targets binned to a 360-class posteriorgram (20 cents/bin). Note MIDI labels are integer-quantised to the nearest semitone.

Augmentation (the core robustness lever). All augmentation is applied on-the-fly at train time (stored features stay clean), and all of it is label-preserving: *gain* (random $-40 \rightarrow 0$ dB, simulating quiet recordings), *spectral-tilt* (random per-band EQ, simulating microphone/room colour), *noise mixing* (FSD-Noise/MUSAN noise at random SNR), and SpecAugment time/frequency masking. Pitch-shift and time-stretch are additionally available offline for technique data, with labels re-derived.

Limitations and bias. (1) The technique and note corpora are VocalSet-only — Western, mostly classically-trained singers — so technique predictions are biased toward that style. (2) Note-pitch labels are semitone-quantised, so expressive vibrato and portamento are inherently ambiguous against a ± 50 -cent tolerance. (3) The SingMOS scale is not calibrated to our user population, and the multi-dimensional expert ratings we inspected are ≈ 0.95 inter-correlated — they carry little independent signal.

5. Model

Architecture. VocalCoachTCN, 7.8M parameters: an input projection (Conv1d) feeds 8 dilated depthwise-separable TCN blocks (kernel 31, exponentially increasing dilation for a wide receptive field well-matched to vibrato periodicity and phrase length), then 4 multi-head self-attention layers (4 heads each), producing frame features ($B, T, 256$) and the five heads of Section 3.

Inputs and outputs. Input ($B, T, 40$) log-mel. Outputs: pitch ($B, T, 360$), VAD ($B, T, 1$), technique ($B, T, 4$), note onset/offset ($B, T, 1$) each, and quality ($B, 1$). At inference the pitch posteriorgram is Viterbi-decoded with full-clip context; notes are formed by peak-picking onsets/offsets and reading per-note MIDI from the

posteriorgram (octave-collapsed, confidence-weighted mean over the span).

Design rationale. We do *not* jointly train one model on everything, and we do *not* fine-tune the whole backbone per task. Fine-tuning the backbone on technique lifts technique average precision (0.70 $\rightarrow$ 0.93) but erodes the loudness-robust VAD (clean voiced-detection rate 87% $\rightarrow$ 63%) — the exact robustness the augmentation bought. The resolution is a *VAD-preserving probe*: train an augmentation-hardened pitch/VAD/note backbone once (gain + spectral-tilt), then attach each new head as a probe on the frozen (or differentially low-learning-rate) backbone. This keeps OOD VDR intact while still reaching state-of-the-art technique accuracy. The deployed system is therefore one backbone plus per-head probes, each adding one forward pass.

Training details. Pitch is a VAD-masked binary cross-entropy over the posteriorgram (Gaussian target, $\sigma = 0.8$ bins). Note onset/offset uses positive-weighted BCE (weight 35) because boundaries are $\sim 2\text{--}3\%$ of frames and unweighted BCE collapses to the majority class. A small octave-confusion penalty on the pitch head was tried for the note head and found not to help; the error is intrinsic (Section 6).

6. Evaluation

Metrics and why they connect to the goal. Because the goal is *robust* coaching on *real* audio, the headline results are the in-distribution versus out-of-distribution (Vocadito, 40 recordings never seen in training). Per head: pitch uses raw pitch accuracy (RPA, ± 50 cents) — does the user see the right pitch?; VAD uses voiced-detection rate (VDR) and voicing F1 (vF1), the hardest part on quiet real audio and what the augmentation targets; technique uses clip accuracy (argmax, matching the SOTA metric) and per-class F1 / macro AP; note uses onset/offset F1 (mir_eval, ± 50 ms) and note-with-pitch F1 (onset ± 50 ms and pitch ± 50 cents), separating timing from pitch-at-note; quality uses Pearson/Spearman correlation with MOS and mean absolute error, because the deployed score is the absolute head output (the ranking head is a training diagnostic, not read at inference).

Headline results (deployed checkpoints) are in Table 3. Per-technique F1 on VocalSet is vibrato 0.94, breathy 0.84, belt 0.79, straight 0.78 (macro AP 0.87): the acoustically most distinctive techniques are detected most reliably.

Discussion. The augmentation result is the clearest win (Figure 4): gain and spectral-tilt lifted OOD VDR from a $\sim 53\%$ baseline to 68% without hurting in-distribution (88% VDR, 99% RPA). RPA barely moves OOD (99 $\rightarrow$ 72%), confirming the residual gap is voicing, not pitch. Technique matches published SOTA. For notes, timing is strong (0.81) but pitch-at-note plateaus at ~ 0.53 : across 27 tuning runs it never es-

Head	Metric	In-dist	OOD
Pitch	RPA	99%	72%
VAD	VDR	88%	68%
VAD	vF1	92%	75%
Technique	clip accuracy	80.8%	—
Note	onset/offset F1	0.81	—
Note	note-with-pitch F1	0.53	—
Quality	Pearson r (MOS)	0.54	—

Table 3: Headline results for the deployed checkpoints. OOD = Vocadito. Pitch and VAD are reported both in-distribution and OOD; the other heads have no OOD test set.

caped a 0.51–0.53 band. We traced this to the backbone posteriorgram — frame-level RPA is 94% and unbiased, but ~18–20% of notes carry a full-semitone octave/harmonic error that no span aggregation removes. An inference-side fix (octave-collapse plus confidence-weighted mean) moved the number from 0.50 to 0.53; the rest is intrinsic. Quality at Pearson 0.54 is a genuine if modest correlation with MOS, and the light pro/amateur ranking term *improved* the absolute score (0.40 → 0.54) rather than fighting it.

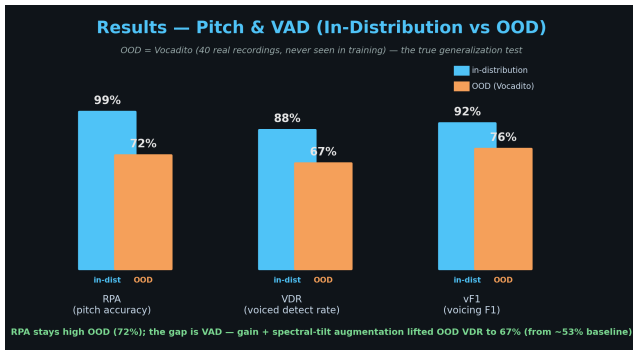


Figure 4: Pitch and VAD, in-distribution versus OOD (Vocadito). Pitch accuracy holds up OOD; gain and spectral-tilt augmentation closed most of the voicing gap.

Two negative results. First, a GTSinger falsetto specialist was trained to add a fifth technique class. On its own labelled test set it output ~0.70 falsetto probability on *both* falsetto (0.704) and non-falsetto (0.679) clips — a 0.025 separation. No threshold distinguishes them; the apparently high F1 was the 71% base rate. We dropped it and ship four trustworthy classes. Second, a nine-dimension expert quality head could not learn independent dimensions because the labels are ≈ 0.95 inter-correlated, only 132 clips, and on a different (1–10) scale; it also corrupted the overall score by conflating it with the “pitch” dimension. We ship the scalar head instead.

Aspect	Detail
Compute	single RTX 4080 Super (16 GB)
Model size	7.8M params (~30 MB fp32)
Mode	offline (non-causal + Viterbi)
Inference	< 1 s per 16 s clip on GPU
Input	WAV/MP3/FLAC, any rate
Deployment	FastAPI + browser UI, local

Table 4: System specifications.

7. System Specifications

The backbone is non-causal and uses a full-clip Viterbi decode, so analysis is post-recording rather than streaming. Technique, note, and quality run as additional forward passes of the shared backbone. Training requires PyTorch; RMVPE is needed only to re-extract pitch labels from raw audio. The UI auto-persists each take (report JSON plus audio) for later replay. A worked example (input and full JSON output) is provided with the code.

8. Limitations and Future Work

Gaps. The note-pitch ceiling (~0.53) is intrinsic to the backbone posteriorgram, not the note head; we exhausted inference-side and light-regularisation fixes, and present note onset/offset F1 (0.81) as the honest headline. Technique is style-biased and four-class; we do not claim falsetto because we could not detect it reliably. Quality is correlational (Pearson 0.54), not a calibrated per-singer 0–100, and the multi-dimensional breakdown we wanted is not supported by available labels. The system is offline only: the non-causal backbone and full-clip decode preclude real-time feedback as built.

Future directions. (1) *Direct per-note pitch head.* Instead of reading note pitch from the frame posteriorgram, pool backbone features over each detected note span and predict MIDI directly with an octave-aware loss. This targets the measured ~18–20% octave-error tail that span aggregation cannot fix — the only lever expected to move note-with-pitch F1 past ~0.53. (2) *Calibrated, streaming quality.* Collect a quality dataset with separable dimensions and a consistent scale, calibrate the 0–100 output to human judgements on amateur practice audio, and train a causal/limited-lookahead backbone so the same heads deliver low-latency feedback during singing, not only after. A third, lighter direction is broadening the technique corpus beyond VocalSet so the detector — and ideally a properly separable falsetto class — generalises across singing styles.

Reproducibility

Code, training scripts, deployment checkpoints, and a worked example are provided with the project repository.

Competing Interests

The author has no competing interests to declare.

AI Usage Statement

An AI coding assistant (Anthropic’s Claude, via the Claude Code CLI) was used during development to assist with code implementation, debugging, refactoring, and to help analyse experimental logs. All research decisions — problem framing, model and training design, choice and interpretation of experiments, and the conclusions drawn (including the decisions to drop the falsetto specialist and the nine-dimension quality head) — were made by the author, who reviewed, tested, and verified all code and results and takes full responsibility for the content of this report.

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